

Hadi Fares

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EDUCATION

American University of Beirut

Bachelor's degree in Mechanical Engineering, GPA: 4.0/4.0

Beirut, Lebanon

Aug. 2023 – May 2027

PUBLICATIONS

- [Publication 1 Title Here] — *Submitted / Under Review.*
- [Publication 2 Title Here] — *Status here.*

EXPERIENCE

GNC Research Assistant - CraneAERO Team

Cranfield University

Jan 2026 – Present

United Kingdom

- Member of CraneAERO, one of 11 global winners of the GoAERO Challenge, developing the CRANE (Cranfield Rapid Aerial Network for Emergency) autonomous eVTOL aircraft for search-and-rescue, disaster relief, and medical evacuation missions.
- Contribute to guidance, navigation, and control (GNC) design for autonomous VTOL flight, supporting safety, mission-readiness, and performance objectives.

Computer Vision & Embedded Systems Intern

Oreyeon

June 2026 – Present

- Developing computer vision and embedded systems solutions, integrating perception pipelines with real-time hardware platforms.

Undergraduate Research Assistant - Aerial Robotics Lab

American University of Beirut

July 2024 – Present

Beirut, Lebanon

- Develop autonomous UAV systems spanning perception, navigation, and flight control, working across multiple advanced research projects in vision-based localization, multi-sensor fusion, and deep learning.
- Built an autonomous firefighting fixed-wing UAV with onboard fire detection and solution dispersal, and a high-speed racing drone currently in testing aiming for a world record.

Undergraduate Research Assistant - AI & Machine Learning Lab

American University of Beirut

July 2024 – Present

Beirut, Lebanon

- Contribute to projects merging drones and aerial robotics with advanced AI, applying large language models (LLMs), neural networks, reinforcement learning (RL), and RLHF to autonomous systems.

Co-Founder & Engineering Lead

AEROClean - Autonomous Window Cleaning Startup

Jan. 2024 – Present

Beirut, Lebanon

- Design and develop autonomous drone systems integrating LiDAR, cameras, and IMU sensors for precise navigation and positioning control.
- Implement flight control algorithms and sensor fusion techniques to enable stable autonomous operation for high-rise building applications.

Head Of JEDIs (Junior Engineering Design Innovator)

AUB Makerspaces

Sep. 2024 – Present

Beirut, Lebanon

- Lead engineering projects with students and professors, specializing in fabrication, electronics, and circuit design using advanced manufacturing tools.
- Manage Red Room operations, coordinate the JEDI team, and organize all makerspace events and workshops.
- Developed carbon fiber-reinforced composites from recycled plastic bottles with optimized 3D printing parameters.

PROJECTS

Real-Time GPS-Denied Drone Navigation System | *Python, PyTorch, FAISS, RLHF, OpenCV*

- Developed visual localization system with autonomous window sizing and correlation-based orientation detection, achieving 97% accuracy and 59 FPS on preprocessed satellite maps.
- Implemented MobileNet-V3 embeddings with FAISS similarity search, validated across 75,000 test frames outperforming transformer methods; journal paper submitted for publication.

Autonomous Balloon-Popping Drone | *Python, OpenCV, YOLO, ROS, AirSim, ArduPilot, Raspberry Pi*

- Developed autonomous drone system using camera and ultrasonic sensors for real-time navigation, target detection, and approach control.
- Implemented YOLO-based red balloon detection with lock-on tracking and autonomous path planning to actuate robotic arm for balloon interception.

JetTracker AI | *Python, PyTorch, LSTM, OpenCV, Reinforcement Learning, Flask*

- Developed real-time jet tracking using LSTM networks and reinforcement learning for path planning, deployed as dockerized web application.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, HTML/CSS, Arduino

Frameworks: PX4 Autopilot, AirSim, ROS, Ubuntu, Gazebo, MATLAB/Simulink, PyTorch, TensorFlow

Machine Learning: RLHF, MobileNet-V3, FAISS, OpenCV, LSTM, CNN, Reinforcement Learning, YOLO

Robotics & Autonomous Systems: Visual Localization, Multi-Sensor Fusion, Path Planning, Flight Control, GPS-Denied Navigation

Software: SolidWorks, Fusion360, ANSYS, Git, Docker, VS Code, AutoCAD, LabVIEW, Flask

Hardware: IMU Systems, GPS, Drone Flight Controllers, Load Cells, PLC Systems, 3D Printing, LiDAR, Radar, Raspberry Pi